Digestion in small intestine

Digestion at the Brush Border

The di and tripeptides formed by the pancreatic enzymes are finally digested by the amino-peptidases and di and tripeptidases present on the luminal side of the intestinal brush border membrane. Leucine aminopeptidase that releases the N-terminal leucine residue and Proline amino peptidase specific for proline at the end of polypeptides, along with the other brush border membrane enzymes, will bring about the complete digestion of proteins.

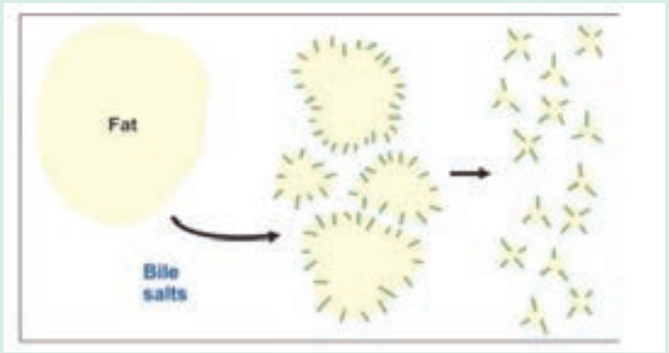

Figure 2.18 Digestion in small intestine

Absorption of amino acids

1. At physiologic temperature, the amino acids and to some extent, the di and tripeptides are absorbed actively by sodium dependent transporters present in the villi. D amino acids produced by bacteria are absorbed passively, while L amino acids are absorbed actively.

At least five brush border transporters exist - Neutral amino acids (uncharged aliphatic and aromatic), basic amino acids (Lys, Arg, Cys, Cys-Cys), acidic amino acids (Asp, Glu), imino acids (Pro, Hydroxyproline) and di- and tripeptides.

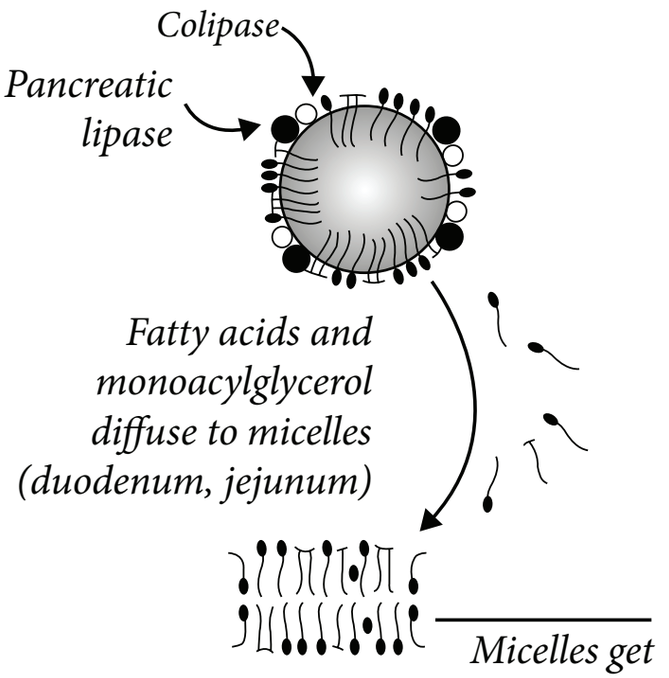

Figure 2.19 Action of pancreatic lipase

Meister Cycle (Gamma Glutamyl Cycle)

Absorption of neutral amino acids is facilitated by the gamma glutamyl cycle. Glutathione (GSH-gamma glutamyl cysteinyl glycine), a tripeptide is involved in this cycle (Figure 2.18) The steps involved in the reaction are:

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